

PLANT REBUILD CENTER

TORQUE FLOW RECEIVING CHECK SHEET

MACHINE MODEL

HD 1500-7

WO NO:

POWER TRAIN SECTION

RECEIVING CHECK SHEET

TORQFLOW

HD 1500 - 7

WO. NO		DOCUMENT NO	
CODE UNIT		PUBLIC DATE	
INSPECTION DATE		REVISION NO	
JOB SITE		PAGE	

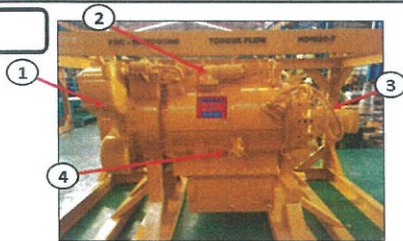
✓ : GOOD CONDITION

X : BAD CONDITION

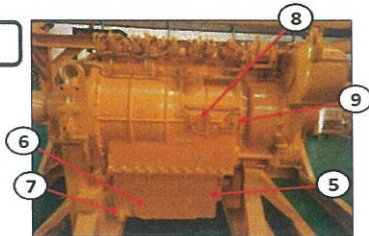
⊗ : NONE

SKETCH DRAWING OR PHOTOES

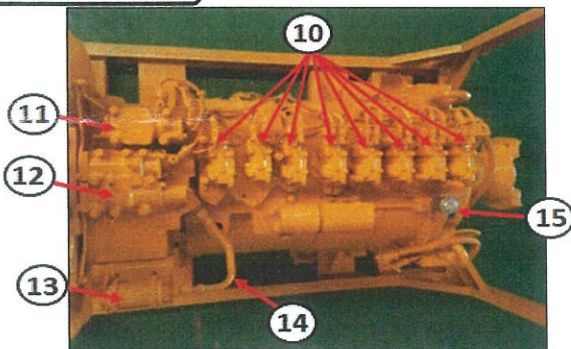
Left Side View



Right Side View

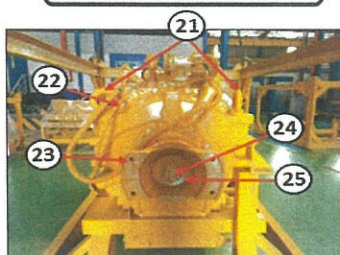
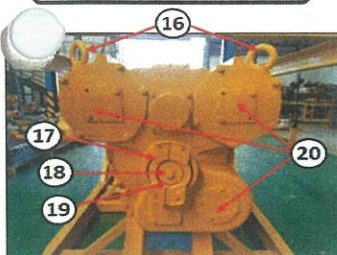


Top Side View



Front Side View

Rear Side View



NOTE : PLEASE MARK "✓" FOLLOWING TO THE ACTUAL CONDITION

OIL LEVEL	EMPTY OR DRAINED	
	PROPER LEVEL	

NO	ITEM	CHECK	REMARK
1	Torque Converter		Comp.S/N:
2	Oil Filter Assy		
3	Wiring Harness		
4	Lubricating Valve Assy		
5	Oil Pan		
6	Magnetic Strainer with Cover		
7	Drain Oil Plug		
8	Revolution Sensor (Front)		
9	Revolution Sensor (Intermediate)		
10	ECMV Assy (8 pcs)		
11	T/C Regulator Valve		
12	Main Relief Valve		
13	Lock Up Accumulator		
14	Lock Up Accumulator Piping		
15	Breather		
16	Eye Bolt Front (2 pcs)		
17	Front Coupling		
18	Bolt Mounting Coupling		
19	Plate		
20	PTO (3pcs)		
21	Eye Bolt Rear (2pcs)		
22	Name Plate Original		
23	Rear Coupling		
24	Bolt Mounting Coupling Rear		
25	Plate		
	Painting		
	Stand		
	Masking		
	Sticker Oil Level		

INSPECTION BY

CHECK BY

CHECK BY

APPROVED BY

QA INSPECTOR

QA COORDINATOR

GL/SPV PROD

PPC

PLANT REBUILD CENTER

TORQUE FLOW VISUAL & MEASUREMENT CHECK SHEET MACHINE MODEL

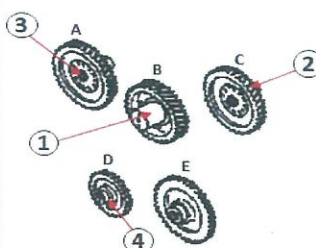
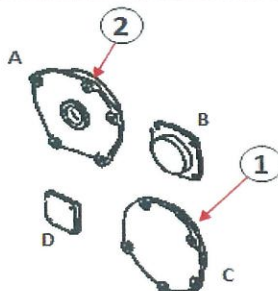
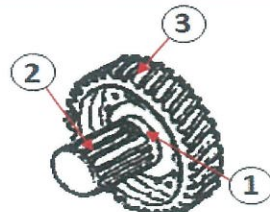
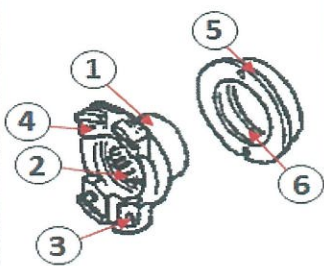
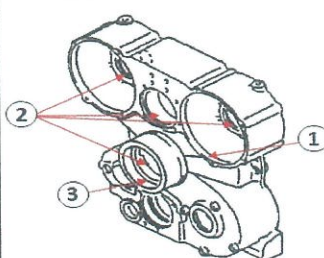
HD 1500

WO NO:

POWERTRAIN SECTION

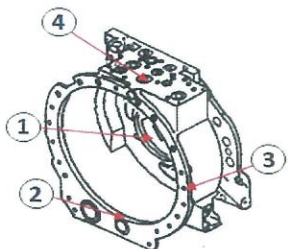
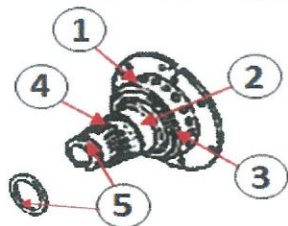
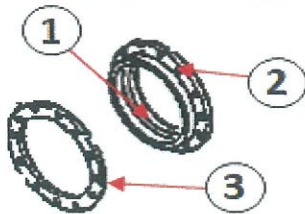
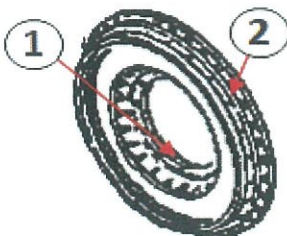
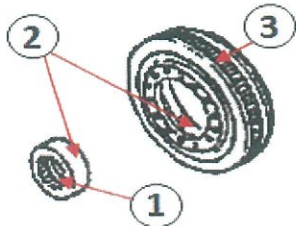
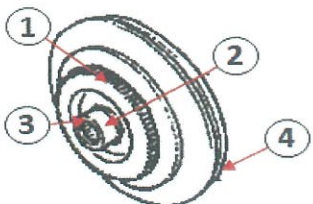
VISUAL & MEASUREMENT CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER		DOCUMENT NO	RC/QR03/PT/05/01
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT CODE		PAGE	1 of 8

PROCESS	START		INSPECTION BY		
	FINISH				

NO	PART NAME	CHECKING POINT	SKECTH DRAWING OR PHOTOES	VISUAL CHECK & MEASUREMENT						DECISION			
				Contens	A	B	C	D	E	U/A	U/R	R/N	
1	Gear a)562-13-44311 b)562-13-44211 c)562-13-44311 d)562-13-44411 e)562-13-44112	1. Bearing fitting portion 2. External gear teeth 3. Internal spline teeth 4. Snap ring groove All round		Contens	A	B	C	D	E				
				Normal						A			
				Crack						B			
				Scoring						C			
				Broken						D			
				Abn wear						E			
				Remark :									
	Cover a)562-13-14341 b)562-13-54450 c)562-13-14351 d)562-13-14421	1. Bearing fitting portion 2. Oring groove All round		Contens	A	B	C	D					
				Normal						A			
				Crack						B			
				Scoring						C			
				Broken						D			
				Abn wear									
				Remark :									
3	Shaft 562-13-51350	1. Bearing fitting portion 2. External spline teeth 3. External gear teeth All round		Contens	Act								
				Normal									
				Crack									
				Scratch									
				Broken									
				Abn wear									
				Remark :									
	Coupling 562-13-51370 Retainer 562-13-53371	1. Seal contact surface Dept : std ≤ 0.1mm Widt : std ≤ 0.5 mm		Contens	Cpng	Rtnr							
				Normal									
		Crack											
		Scratch											
		Broken											
		Abn wear											
		Remark :											
		Seal contact size											
		Actual											
Cpng													
Rtnr													
5	Housing 562-13-53132	1. Oring contact surface 2. Bearing fitting portion 3. Retainer fitting surface Collar contact surface		Contens	Act								
				Normal									
		Crack											
		Scratch											
		Broken											
		Abn wear											
		Remark :											
		Collar contact surface											
		Actual											

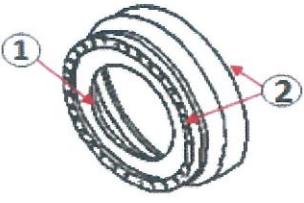
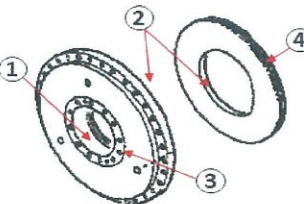
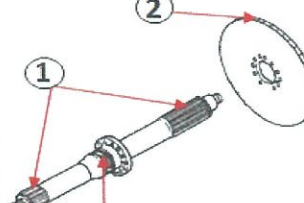
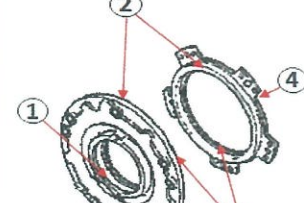
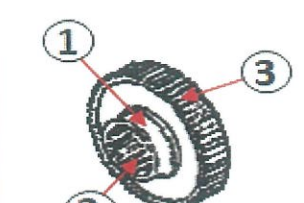
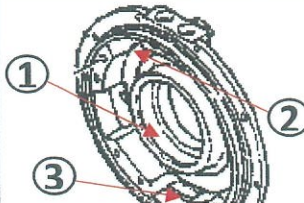
VISUAL & MEASUREMENT CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER		DOCUMENT NO	RC/QR03/PT/05/01
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NO	PART NAME	CHECKING POINT	SKECTH DRAWING OR PHOTOES	VISUAL CHECK & MEASUREMENT		DECISION																												
						U/A	U/R	R/N																										
6	Housing 562-13-53110	1. Shaft fitting surface 2. Oring contact surface 3. Thread fitting portion 4. Oring groove All round		<table><tr><td>Contens</td><td>Act</td></tr><tr><td>Normal</td><td></td></tr><tr><td>Crack</td><td></td></tr><tr><td>Scratch</td><td></td></tr><tr><td>Broken</td><td></td></tr><tr><td>Abn wear</td><td></td></tr></table>	Contens	Act	Normal		Crack		Scratch		Broken		Abn wear		Remark :	<table><tr><td></td><td></td><td></td></tr></table>																
Contens	Act																																	
Normal																																		
Crack																																		
Scratch																																		
Broken																																		
Abn wear																																		
7	Shaft 562-13-42520	1. Housing fitting portion 2. Bearing fitting portion 3. Ring groove 4. External spline teeth 5. Thread fitting portion All round		<table><tr><td>Contens</td><td>Shft</td><td>Nut</td></tr><tr><td>Normal</td><td></td><td></td></tr><tr><td>Crack</td><td></td><td></td></tr><tr><td>Scratch</td><td></td><td></td></tr><tr><td>Broken</td><td></td><td></td></tr><tr><td>Abn wear</td><td></td><td></td></tr></table>	Contens	Shft	Nut	Normal			Crack			Scratch			Broken			Abn wear			Remark :	<table><tr><td>Shft</td><td></td><td></td><td></td></tr><tr><td>Nut</td><td></td><td></td><td></td></tr></table>			Shft				Nut			
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Broken																																		
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Shft																																		
Nut																																		
8	Plate 562-13-11192 Retainer 562-13-11145	1. Bearing fitting portion 2. Thread fitting portion 3. Pump contact surface All round		<table><tr><td>Contens</td><td>Plt</td><td>Rtnr</td></tr><tr><td>Normal</td><td></td><td></td></tr><tr><td>Crack</td><td></td><td></td></tr><tr><td>Scratch</td><td></td><td></td></tr><tr><td>Broken</td><td></td><td></td></tr><tr><td>Abn wear</td><td></td><td></td></tr></table>	Contens	Plt	Rtnr	Normal			Crack			Scratch			Broken			Abn wear			Remark :	<table><tr><td>Plt</td><td></td><td></td><td></td></tr><tr><td>Rtnr</td><td></td><td></td><td></td></tr></table>			Plt				Rtnr			
Contens	Plt	Rtnr																																
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Crack																																		
Scratch																																		
Broken																																		
Abn wear																																		
Plt																																		
Rtnr																																		
9	Pump 562-13-41121	1. Retainer contact surface 2. Sudu of pump All round		<table><tr><td>Contens</td><td>Act</td></tr><tr><td>Normal</td><td></td></tr><tr><td>Crack</td><td></td></tr><tr><td>Scratch</td><td></td></tr><tr><td>Broken</td><td></td></tr><tr><td>Abn wear</td><td></td></tr></table>	Contens	Act	Normal		Crack		Scratch		Broken		Abn wear		Remark :	<table><tr><td></td><td></td><td></td></tr></table>																
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Crack																																		
Scratch																																		
Broken																																		
Abn wear																																		
10	Race 562-13-12132 Stator 562-13-42110	1. Internal spline teeth 2. Free whell fitting portion 3. Sudu of stator All round		<table><tr><td>Contens</td><td>Race</td><td>Sstr</td></tr><tr><td>Normal</td><td></td><td></td></tr><tr><td>Crack</td><td></td><td></td></tr><tr><td>Scratch</td><td></td><td></td></tr><tr><td>Broken</td><td></td><td></td></tr><tr><td>Abn wear</td><td></td><td></td></tr></table>	Contens	Race	Sstr	Normal			Crack			Scratch			Broken			Abn wear			Remark :	<table><tr><td>Race</td><td></td><td></td><td></td></tr><tr><td>Sstr</td><td></td><td></td><td></td></tr></table>			Race				Sstr			
Contens	Race	Sstr																																
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Crack																																		
Scratch																																		
Broken																																		
Abn wear																																		
Race																																		
Sstr																																		
11	Turbin 562-13-51511	1. External spline teeth 2. Bearing fitting portion 3. Snap ring groove 4. Sudu of turbin All round		<table><tr><td>Contens</td><td>Act</td></tr><tr><td>Normal</td><td></td></tr><tr><td>Crack</td><td></td></tr><tr><td>Scratch</td><td></td></tr><tr><td>Broken</td><td></td></tr><tr><td>Abn wear</td><td></td></tr></table>	Contens	Act	Normal		Crack		Scratch		Broken		Abn wear		Remark :	<table><tr><td></td><td></td><td></td></tr></table>																
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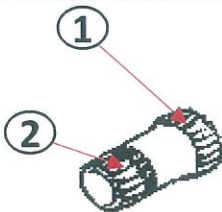
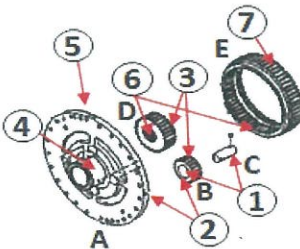
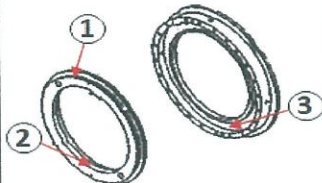
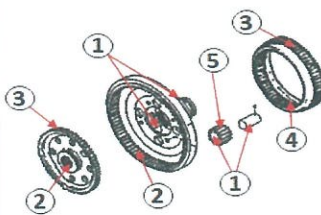
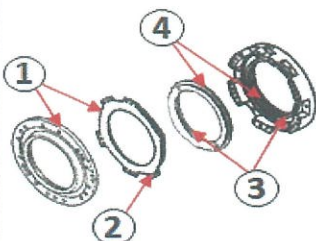
VISUAL & MEASUREMENT CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER		DOCUMENT NO	RC/QR03/PT/05/01
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PROCESS	START		INSPECTION BY	
	FINISH			

NO	PART NAME	CHECKING POINT	SKECTH DRAWING OR PHOTOES	VISUAL CHECK & MEASUREMENT		DECISION																										
						U/A	U/R	R/N																								
12	Case 562-13-51210	1. Disc contact surface 2. Thread fitting portion All round		<table><tr><td>Contens</td><td>Act</td></tr><tr><td>Normal</td><td></td></tr><tr><td>Crack</td><td></td></tr><tr><td>Scratch</td><td></td></tr><tr><td>Broken</td><td></td></tr><tr><td>Abn wear</td><td></td></tr></table>	Contens	Act	Normal		Crack		Scratch		Broken		Abn wear		Remark :	<table><tr><td></td><td></td><td></td></tr></table>														
Contens	Act																															
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Crack																																
Scratch																																
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Abn wear																																
	Piston 562-13-57210 Housing 562-13-57111	1. Bearing fitting portion 2. Seal contact surface 3. Thread fitting portion 4. Seal groove All round		<table><tr><td>Contens</td><td>Pstn</td><td>Hsng</td></tr><tr><td>Normal</td><td></td><td></td></tr><tr><td>Crack</td><td></td><td></td></tr><tr><td>Scratch</td><td></td><td></td></tr><tr><td>Broken</td><td></td><td></td></tr><tr><td>Abn wear</td><td></td><td></td></tr></table>	Contens	Pstn	Hsng	Normal			Crack			Scratch			Broken			Abn wear			Remark :	<table><tr><td>Pstn</td><td></td><td></td><td></td></tr><tr><td>Hsng</td><td></td><td></td><td></td></tr></table>	Pstn				Hsng			
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14	Plate 562-15-22980 Shaft 562-15-22110	1. External spline teeth 2. External Gear teeth 3. Ring Groove All round		<table><tr><td>Contens</td><td>Plt</td><td>Shft</td></tr><tr><td>Normal</td><td></td><td></td></tr><tr><td>Crack</td><td></td><td></td></tr><tr><td>Scratch</td><td></td><td></td></tr><tr><td>Broken</td><td></td><td></td></tr><tr><td>Abn wear</td><td></td><td></td></tr></table>	Contens	Plt	Shft	Normal			Crack			Scratch			Broken			Abn wear			Remark :	<table><tr><td>Plt</td><td></td><td></td><td></td></tr><tr><td>Shft</td><td></td><td></td><td></td></tr></table>	Plt				Shft			
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15	Housing 562-15-31190 Piston 562-15-32780	1. Bearing fitting portion 2. Seal ring groove 3. Seal contact surface 4. Disc contact surface All round		<table><tr><td>Contens</td><td>Hsng</td><td>Pstn</td></tr><tr><td>Normal</td><td></td><td></td></tr><tr><td>Crack</td><td></td><td></td></tr><tr><td>Scratch</td><td></td><td></td></tr><tr><td>Broken</td><td></td><td></td></tr><tr><td>Abn wear</td><td></td><td></td></tr></table>	Contens	Hsng	Pstn	Normal			Crack			Scratch			Broken			Abn wear			Remark :	<table><tr><td>Hsng</td><td></td><td></td><td></td></tr><tr><td>Pstn</td><td></td><td></td><td></td></tr></table>	Hsng				Pstn			
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Crack																																
Scratch																																
Broken																																
Abn wear																																
Hsng																																
Pstn																																
16	Hub 562-15-52270	1. Bearing contact surface 2. Internal spline teeth 3. External Spline teeth All round		<table><tr><td>Contens</td><td>Act</td></tr><tr><td>Normal</td><td></td></tr><tr><td>Crack</td><td></td></tr><tr><td>Scratch</td><td></td></tr><tr><td>Broken</td><td></td></tr><tr><td>Abn wear</td><td></td></tr></table>	Contens	Act	Normal		Crack		Scratch		Broken		Abn wear		Remark :	<table><tr><td></td><td></td><td></td></tr></table>														
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Normal																																
Crack																																
Scratch																																
Broken																																
Abn wear																																
17	Housing 562-15-31180	1. Bearing fitting portion 2. Disc contact surface 3. Spring fitting portion All round		<table><tr><td>Contens</td><td>Act</td></tr><tr><td>Normal</td><td></td></tr><tr><td>Crack</td><td></td></tr><tr><td>Scratch</td><td></td></tr><tr><td>Broken</td><td></td></tr><tr><td>Abn wear</td><td></td></tr></table>	Contens	Act	Normal		Crack		Scratch		Broken		Abn wear		Remark :	<table><tr><td></td><td></td><td></td></tr></table>														
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PROCESS	START		INSPECTION BY		
	FINISH				

NO	PART NAME	CHECKING POINT	SKEETH DRAWING OR PHOTOES	VISUAL CHECK & MEASUREMENT				DECISION				
				Contens	Act	U/A	U/R	R/N				
18	Gear 562-15-12211	1. External gear teeth 2. External spline teeth All round		Contens	Act	Remark :						
		Normal										
		Crack										
		Scratch										
		Broken										
				Abn wear								
19	Carrier 562-15-22310 Gear 562-15-12411 Shaft 562-15-12511 Gear 562-15-22220 Ring gear 562-15-22650	1. Bearing fitting portion 2. Trust washer contact surface 3. External gear teeth 4. Matting face 5. Disc contact surface 6. Internal spline 7. External spline teeth All round		Contens	A	B	C	D	E			
				Normal						A		
				Crack						B		
				Scratch						C		
				Broken						D		
				Abn wear						E		
				Remark :								
20	Piston 562-15-22770 Plate 562-15-22840	1. Seal groove 2. Seal contact surface 3. Thread fitting portion All round		Contens	Plte	Pstn	Remark :			Plte Pstn		
		Normal										
		Crack										
		Scratch										
		Broken										
				Abn wear								
21	Ring gear 562-15-22640 Carrier 562-15-22321 Gear 562-15-12420 Hub 562-15-22280 Shaft 562-15-32520	1. Bearing fitting portion 2. Internal spline teeth 3. External spline teeth 4. Internal gear teeth 5. External gear teeth All round		Contens	gear	Carr	Gear	Hub	Shft			
				Normal						gear		
				Crack						Carr		
				Scratch						Gear		
				Broken						Hub		
				Abn wear						Shft		
				Remark :								
22	Plate 562-15-22970 Plate 562-15-21171 Housing 562-15-21160 Piston 562-15-22960	1. Disc contact surface 2. Spring fitting portion 3. Seal contact surface 4. Seal groove All round		Contens	Plte	Plte	Hsng	Pstn	Plte Plte Hsng Pstn			
				Normal								
				Crack								
				Scratch								
				Broken								
				Abn wear								
				Remark :								

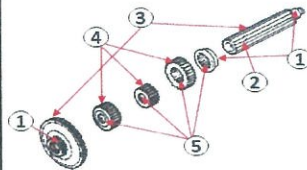
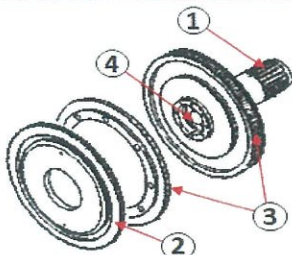
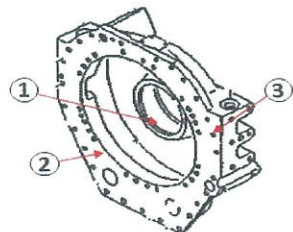
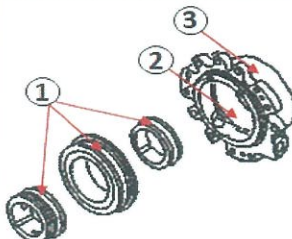
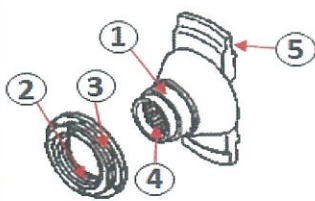
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	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT CODE		PAGE	5 of 8

PROCESS	START		INSPECTION BY		
	FINISH				

NO	PART NAME	CHECKING POINT	SKECTH DRAWING OR PHOTOS	VISUAL CHECK & MEASUREMENT					DECISION		
				Contens	Hsng	Plte	Pstn	Hsng	U/A	U/R	R/N
23	Housing 562-15-51120 Plate 562-15-22740 Piston 562-15-32760 Housing 562-15-51150	1. Disc contact surface 2. Seal contact surface 3. Seal ring groove 4. Spring fitting portion All round		Contens	Hsng	Plte	Pstn	Hsng			
				Normal					Hsng		
				Crack					Plte		
				Scratch					Pstn		
				Broken					Hsng		
				Abn wear							
				Remark :							
	Carrier 562-15-22330 Gear 562-15-22430 Shaft 562-15-32530 Ring gear 562-15-32630	1. Bearing fitting portion 2. External gear teeth 3. External spline teeth 4. Trust washer contact surface 5. Eeternal spline 6. Internal Gear teeth All round		Contens	Carr	Gear	Shft	Gear			
				Normal					Carr		
				Crack					Gear		
				Scratch					Shft		
				Broken					Gear		
				Abn wear							
				Remark :							
25	Housing 562-15-21141 Piston 562-15-22752 Plate 562-15-22740	1. Seal contact surface 2. Seal groove 3. Disc contact surface 4. Spring groove All round		Contens	Hsng	Pstn	Plt				
				Normal					Hsng		
				Crack					Pstn		
				Scratch					Plt		
				Broken							
				Abn wear							
				Remark :							
	Housing 562-15-21130 Piston 562-15-22910	1. Disc contact surface 2. Seal contact surface 3. spring groove All round		Contens	Hsng	Pstn					
				Normal					Hsng		
				Crack					Pstn		
				Scratch							
				Broken							
				Abn wear							
				Remark :							
27	Carrier 562-15-22341 Ring gear 562-15-32610 562-15-52620 Gear 562-15-32461 562-15-12440 562-15-22451 Shaft 562-15-32541 562-15-12551	1. Bearing fitting portion 2. External gear teeth 3. Internal spline teeth 4. External spline teeth 5. Internal gear teeth 6. Trust washer contact surface		Contens	Crr	Ring	Gear	Shft			
				Normal					Crr		
				Crack					Ring		
				Scratch					Gear		
				Broken					Shft		
				Abn wear							
				Remark :							

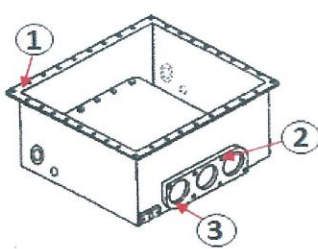
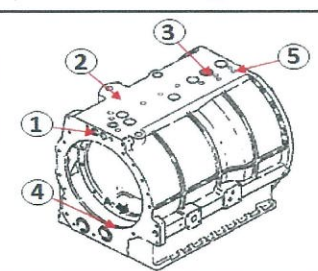
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	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
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PROCESS	START		INSPECTION BY		
	FINISH				

NO	PART NAME	CHECKING POINT	SKECTH DRAWING OR PHOTOES	VISUAL CHECK & MEASUREMENT					DECISION		
									U/A	U/R	R/N
28	Hub 562-15-22290 Gear 562-15-22230 562-15-22240 562-15-32251 Spacer 62-15-22170 Shaft 562-15-22120	1. Bearing fitting portion 2. Matting face 3. External spline teeth 4. External gear teeth 5. Internal spline teeth All round		Contens	Hub	Gear	Spcr	Shft			
	Normal						Hub				
	Crack						Gear				
	Scratch						Spcr				
	Broken						Shft				
	Abn wear										
	Remark :										
29	Shaft 562-15-22130 Gear 562-15-22140 Piston 562-15-22730	1. External spline teeth 2. Disc contact surface 3. External Gear teeth 4. Bearing fitting portion All round		Contens	Shft	Gear	Pstn				
	Normal						Shft				
	Crack						Gear				
	Scratch						Pstn				
	Broken						Remark :				
	Abn wear										
30	Case 562-15-33110	1. Bearing fitting portion 2. Disc contact surface 3. Thread fitting portion All round		Contens	Act						
	Normal		Remark :								
	Crack										
	Scratch										
	Broken										
	Abn wear										
31	Collar 562-15-13140 562-15-13251 Cover 562-15-23120 Worm 562-15-13411	1. Collar contact surface 2. Bearing contact surface 3. Oring groove All round		Contens	Collr	Collr	Cvr	Wrm			
	Normal						Collr				
	Crack						Collr				
	Scratch						Cvr				
	Broken						Wrm				
	Abn wear										
	Remark :										
32	Cover 562-15-33130 Coupling 562-15-23311	1. Seal contact surface 2. Seal fitting portion 3. Oring groove 4. Internal spline teeth 5. Matting face All round		Contens	Hsng	Pstn					
	Normal						Hsng				
	Crack						Pstn				
	Scratch						Remark :				
	Broken										
	Abn wear										

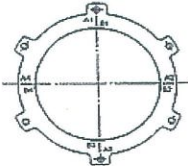
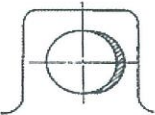
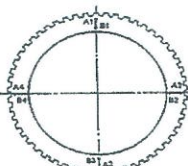
VISUAL & MEASUREMENT CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER		DOCUMENT NO	RC/QR03/PT/05/01
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
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PROCESS	START		INSPECTION BY		
	FINISH				

NO	PART NAME	CHECKING POINT	SKEPTH DRAWING OR PHOTOES	VISUAL CHECK & MEASUREMENT	DECISION																																																													
					U/A	U/R	R/N																																																											
33	Oil pan 562-15-31551	1. Mating face 2. Thread fitting portion 3. Oring groove All round		<table><tr><td>Contens</td><td>Act</td></tr><tr><td>Normal</td><td></td></tr><tr><td>Crack</td><td></td></tr><tr><td>Scratch</td><td></td></tr><tr><td>Broken</td><td></td></tr><tr><td>Abn wear</td><td></td></tr></table> Remark :	Contens	Act	Normal		Crack		Scratch		Broken		Abn wear																																																			
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Abn wear																																																																		
34	Case 562-15-31110	1. Oil galery 2. Matting face 3. Oring groove 4. Oring contact surface 5. Treath fitting portion All round		<table><tr><td>Contens</td><td>Act</td></tr><tr><td>Normal</td><td></td></tr><tr><td>Crack</td><td></td></tr><tr><td>Scratch</td><td></td></tr><tr><td>Broken</td><td></td></tr><tr><td>Abn wear</td><td></td></tr></table> Remark :	Contens	Act	Normal		Crack		Scratch		Broken		Abn wear																																																			
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35	Spring	 <table><thead><tr><th rowspan="3">No.</th><th rowspan="3">Description</th><th colspan="2">Specifications</th></tr><tr><th>Standard</th><th>Limit</th></tr><tr><th>Free Length</th><th>Free Length</th></tr></thead><tbody><tr><td>1</td><td>No. 1 (M) clutch springs (10 ea.)</td><td>91.0 mm (3.58 in.)</td><td>85.5 mm (3.37 in.)</td></tr><tr><td>2</td><td>No. 3 (L) clutch springs (12 ea.)</td><td>91 mm (3.58 in.)</td><td>85.5 mm (3.37 in.)</td></tr><tr><td>3</td><td>No. 4 (2nd) clutch springs (12 ea.)</td><td>91.0 mm (3.58 in.)</td><td>85.5 mm (3.37 in.)</td></tr><tr><td>4</td><td>No. 5 (1st) clutch springs (12 ea.)</td><td>91 mm (3.58 in.)</td><td>85.5 mm (3.37 in.)</td></tr><tr><td>5</td><td>No. 6 (R) clutch springs (12 ea.)</td><td>91 mm (3.58 in.)</td><td>85.5 mm (3.37 in.)</td></tr></tbody></table> Remark :	No.	Description	Specifications		Standard	Limit	Free Length	Free Length	1	No. 1 (M) clutch springs (10 ea.)	91.0 mm (3.58 in.)	85.5 mm (3.37 in.)	2	No. 3 (L) clutch springs (12 ea.)	91 mm (3.58 in.)	85.5 mm (3.37 in.)	3	No. 4 (2nd) clutch springs (12 ea.)	91.0 mm (3.58 in.)	85.5 mm (3.37 in.)	4	No. 5 (1st) clutch springs (12 ea.)	91 mm (3.58 in.)	85.5 mm (3.37 in.)	5	No. 6 (R) clutch springs (12 ea.)	91 mm (3.58 in.)	85.5 mm (3.37 in.)	<table><tr><td>M</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>L</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>2nd</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1st</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	M							L							2nd							1st							R						
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36	Pin	1. Matting face All round		<table><tr><td>Contens</td><td>Act</td></tr><tr><td>Normal</td><td></td></tr><tr><td>Crack</td><td></td></tr><tr><td>Scratch</td><td></td></tr><tr><td>Broken</td><td></td></tr><tr><td>Abn wear</td><td></td></tr></table> Remark :	Contens	Act	Normal		Crack		Scratch		Broken		Abn wear																																																			
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VISUAL & MEASUREMENT CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER		DOCUMENT NO	RC/QR03/PT/05/01
	COMP.MODEL		PUBLIC DATE	
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PROCESS	START		INSPECTION BY		
	FINISH				

NO	PART NAME	CHECKING POINT	SKEETH DRAWING OR PHOTOO	VISUAL CHECK & MEASUREMENT	DECISION																												
					U/A	U/R	R/N																										
37	Plate	• Plate wear  • Plate torque pin hole wear and fatigue 	• Plate wear  • Wear on wave spring contact surface 	<table border="1"> <thead> <tr> <th rowspan="2">No</th> <th rowspan="2">Clutch</th> <th colspan="2">Specification</th> </tr> <tr> <th>Standard</th> <th>Tolerance</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>No. 1 clutch (Middle) Single plate thickness:</td> <td>4.3 mm (0.169 in.)</td> <td>±0.1 mm (±0.004 in.)</td> </tr> <tr> <td>2</td> <td>No. 2 clutch (High) Single plate thickness:</td> <td>5.2 mm (0.205 in.)</td> <td>±0.1 mm (±0.004 in.)</td> </tr> <tr> <td>3</td> <td>No. 3 clutch (Low) Single plate thickness:</td> <td>7.0 mm (0.276 in.)</td> <td>±0.1 mm (±0.004 in.)</td> </tr> <tr> <td>4</td> <td>No. 4, 5 & 6 clutch (2nd, 1st, Reverse) Single plate thickness:</td> <td>5.8 mm (0.228 in.)</td> <td>±0.1 mm (±0.004 in.)</td> </tr> <tr> <td>5</td> <td>No. 7 clutch (3rd) Single plate thickness:</td> <td>5.2 mm (0.205 in.)</td> <td>±0.1 mm (±0.004 in.)</td> </tr> </tbody> </table>	No	Clutch	Specification		Standard	Tolerance	1	No. 1 clutch (Middle) Single plate thickness:	4.3 mm (0.169 in.)	±0.1 mm (±0.004 in.)	2	No. 2 clutch (High) Single plate thickness:	5.2 mm (0.205 in.)	±0.1 mm (±0.004 in.)	3	No. 3 clutch (Low) Single plate thickness:	7.0 mm (0.276 in.)	±0.1 mm (±0.004 in.)	4	No. 4, 5 & 6 clutch (2nd, 1st, Reverse) Single plate thickness:	5.8 mm (0.228 in.)	±0.1 mm (±0.004 in.)	5	No. 7 clutch (3rd) Single plate thickness:	5.2 mm (0.205 in.)	±0.1 mm (±0.004 in.)			
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					1ST																												
					R																												
					3RD																												

Note

INSPECTION BY

MECHANIC

APPROVED BY

GL /QA

REV. 00



PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

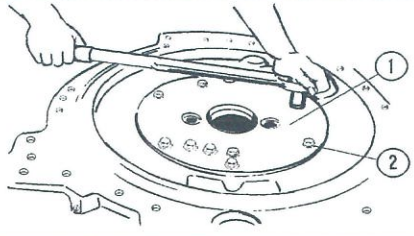
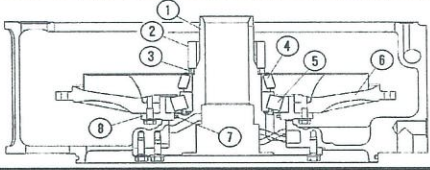
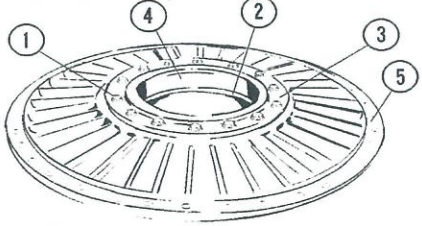
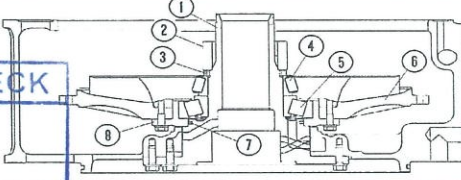
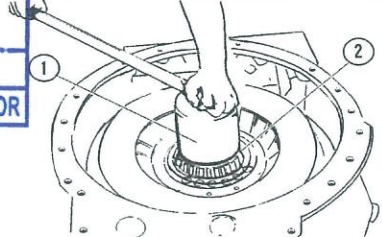
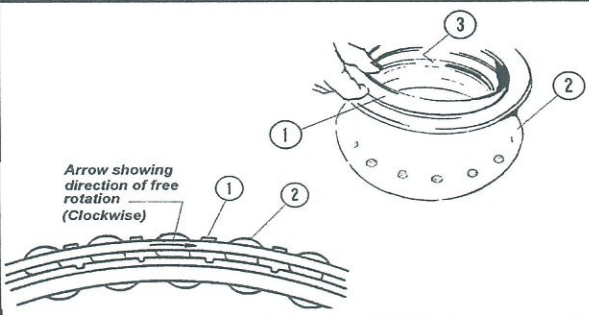
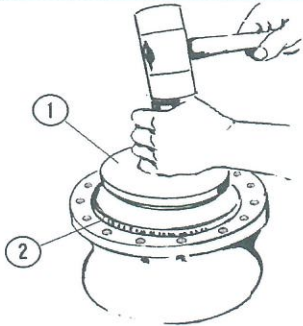
TORQUE CONVERTER ASSEMBLY CHECK SHEET

MACHINE MODEL

HD 1500-7

WO NO:

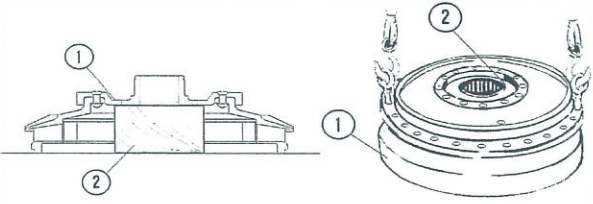
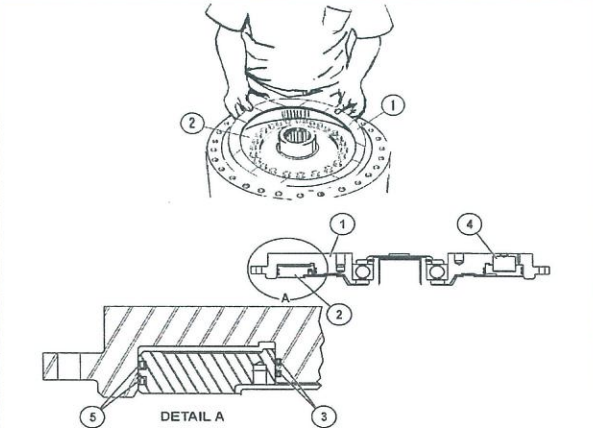
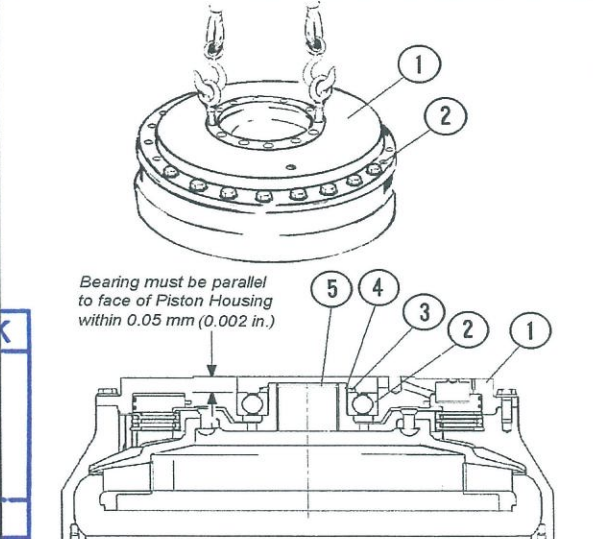
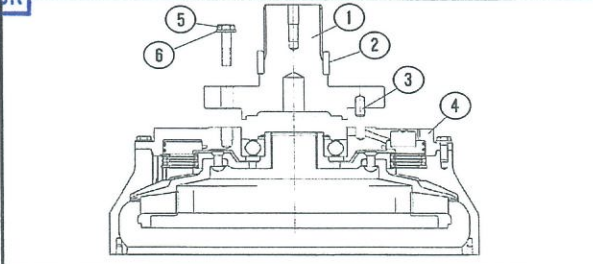
POWER TRAIN SECTION

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO
TORQUE CONVERTER		COMP.MODEL	PUBLIC DATE
HD 1500 - 7		COMP.S/N	REVISION NO
		UNIT MODEL	PAGE
PROCESS	START FINISH	ASSEMBLY BY	
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH GL/QA SKETCH DRAWING OR PHOTHOES
1	SHAFT INSTALLATION	IMPORTANT Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches. Apply thread lock • Install the cap screws & washers. screws (2) Tightening bolt Standard: 11 ± 1.2 Kgm Actual:	
2	PUMP ASSEMBLY	• Press the outer race of bearing (5) into retainer (8).	
3	PUMP SUB ASSEMBLY	• Install retainer (3) and tapped plate (2) onto the pump Apply thread lock • Install the cap screws and washers. Tightening bolt Standard: 6.6 ± 0.7 Kgm Actual:	
4	PUMP ASSEMBLY	• Install seal ring (7) on shaft (1). Coat the seal ring with petroleum jelly. Carefully lower the pump sub-assembly onto the shaft.	
5	TORQUEING SHAFT NUT	• Use special tool Apply thread lock Tightening nut Standard: 46.6 ± 2.4 Kgm Actual:	
6	BEARING RACE AND FREE-WHEEL INSTALLATION	• Install plate (3, Figure 8-48) into the bore of race (2). Place bearing race (1) in dry ice until chilled. Use 19.6 - 29.4 kN (2 - 3 tons) force to press bearing outer race (1) into the bore of race (2). • Turn the race over with the tapped holes positioned at the top. • Position the free-wheel in the bore of the race with the arrow pointing clockwise.	
7	FREE-WHEEL INSTALLATION	• Use a mallet and installation tool (1) to tap the free-wheel into the bore of the race. Use caution when installing. Damage to sprags or drag springs may result with too much force.	

RANDOM ITEM CHECK

Name : QA INSPECTOR
Name : PROD. SUPERVISOR

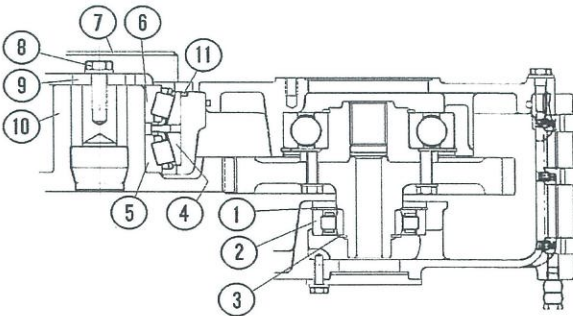
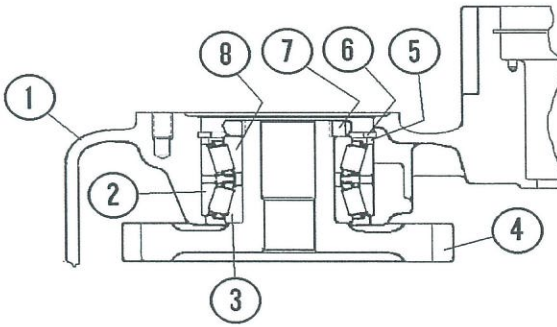
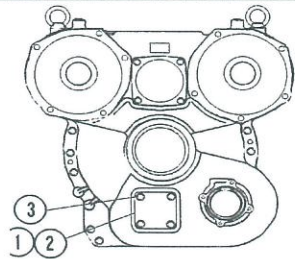
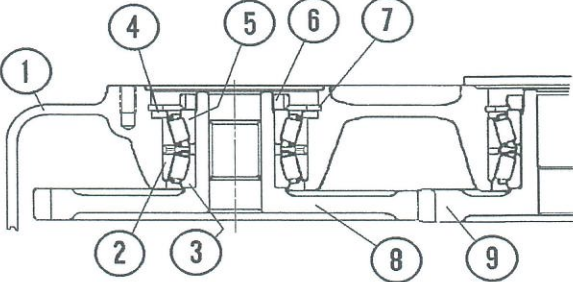
ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO								
TORQUE CONVERTER		COMP.MODEL	PUBLIC DATE								
HD 1500 - 7		COMP.S/N	REVISION NO								
		UNIT MODEL	PAGE								
PROCESS	START FINISH	ASSEMBLY BY									
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH GL/QA SKETCH DRAWING OR PHOTHOES								
8	STATOR INSTALLATION	Place bearing cup (4) in dry ice to chill. Use 19.6 - 29.4 kN (2 - 3 tons) force to press the bearing cup into retainer (3). Verify the cup is properly seated with no between the cup and the shoulder.clearance - Assemble stator (1) retainer (3) and race (10). Apply thread lock - Install the cap screws and the washers. Tightening bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>11 ± 1.2 Kgm</td> <td></td> </tr> </table> - Apply power line oil to the one-way clutch inner race. - Lower stator assembly (1) over the shaft. Position the stator into rear housing (2) while rotating clockwise. - After the stator is seated onto the bearing, rotate the stator. Confirm the stator will turn clockwise, but not counterclockwise.	Standard	Actual	11 ± 1.2 Kgm						
Standard	Actual										
11 ± 1.2 Kgm											
9	STATOR BEARING PRE-LOAD ADJUSTMENT	- Rotate the stator 15 to 20 times. Tap the stator two or three times using a soft mallet. method of checking rolling resistance. - Hook a 10 kg (20 lb) spring scale to the other end of the rope. - Pull the spring scale smoothly. Read and record the force required to start the stator rotating. The front stator bearing cone is not installed. Take several measurements and record the average value. RANDOM ITEM CHECK Name : QA INSPECTOR PROD. SUPERVISOR - Install spacer (9) and bearing cone (5). Coat the bearing with power line oil and rotate the stator several revolutions. - Install nut (8). Use special tool Tightening nut <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>36.7 ± 2.4 Kgm</td> <td></td> </tr> </table> - Loosen nut (8) about 60 degrees. Install lock plate (7) and outer nut (6). Adjust the pre-load by loosening and tightening the stator bearing retainer nuts until the proper tangential force is achieved. The proper tangential force required to rotate the one-way clutch is 3.5 to 4.0 kg (7.7 to 8.8 lbs.) Tangential force = Force required to rotate one-way clutch + 2.2 to 3.6 kg (4.9 to 7.9 lbs.). - After completing the preload adjustment, bend locking plate (7) against the flats of nuts (6 & 8). - Tighten two stator retainer cap screws (2) Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>11 ± 1.2 Kgm</td> <td></td> </tr> </table> - Confirm that all rollers of the stator bearing (5) rotate. Verify that a 0.05 mm (0.002 in.) feeler gauge cannot be inserted between the end of the roller and the lip on the outer race.	Standard	Actual	36.7 ± 2.4 Kgm		Standard	Actual	11 ± 1.2 Kgm		
Standard	Actual										
36.7 ± 2.4 Kgm											
Standard	Actual										
11 ± 1.2 Kgm											

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PROCESS		START		ASSEMBLY BY									
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NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES								
10	CASE INSTALLATION	- Reposition the torque converter with the axis of the shaft vertical - Place the turbine on a work bench. Support the turbine on a wood block - Attach a lifting sling and a hoist to case (1). Lower the case over turbine (2).											
11	CLUTCH ASSEMBLY AND PISTON INSTALLATION	- Install a lock-up clutch disc (1) and then a plate (2). <table border="1"> <tr> <td>Standard disc</td> <td>Actual</td> </tr> <tr> <td>3 pcs</td> <td></td> </tr> </table> <table border="1"> <tr> <td>Standard plate</td> <td>Actual</td> </tr> <tr> <td>2 pcs</td> <td></td> </tr> </table> - Install piston seal rings (5) in the piston grooves as shown (cup positioned toward outside). Coat the piston seal ring with petroleum jelly. - Install piston seal rings (3) into the groove in housing (1). Coat the piston seal ring with petroleum jelly.	Standard disc	Actual	3 pcs		Standard plate	Actual	2 pcs				
Standard disc	Actual												
3 pcs													
Standard plate	Actual												
2 pcs													
12	BEARING AND CLUTCH HOUSING INSTALLATION	- Apply thread lock to cap screws (2). Install the cap screws and washers and tighten Tightening cap screw <table border="1"> <tr> <td>Standard</td> <td>Actual</td> </tr> <tr> <td>11 ± 1.2 Kgm</td> <td></td> </tr> </table> - Press fit bearing (2) into housing bore. Verify the bearing is installed parallel to the face of piston housing (1) within 0.05 mm (0.002 in.). Install spacer (3) and snap rings (4 & 5). - Press dowel pins (3) into the four holes in bottom of input shaft (1). - Press bearing inner race (2) onto the shaft until seated against the shoulder. - Align the dowel pins in the shaft with the holes in piston housing (4). Tap the shaft until seated on the housing.	Standard	Actual	11 ± 1.2 Kgm				 Bearing must be parallel to face of Piston Housing within 0.05 mm (0.002 in.)				
Standard	Actual												
11 ± 1.2 Kgm													
13	SHAFT INSTALLATION	Apply thread lock Install cap screws (5) and washers (6). Tightening cap screw <table border="1"> <tr> <td>Standard</td> <td>Actual</td> </tr> <tr> <td>29.2 ± 1.8 Kgm</td> <td></td> </tr> </table> After assembly, check the turbine for smooth rotation.	Standard	Actual	29.2 ± 1.8 Kgm								
Standard	Actual												
29.2 ± 1.8 Kgm													

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	FINISH												
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTOES								
14	PTO UPPER PUMP DRIVE & IDLER GEAR ASSEMBLY	- Press fit bearing outer race (2) for the right hand pump drive into the front housing.Install snap ring (1). - Install outer bearing race (4) for the upper idler gear in the front housing bore. - Press fit half of inner race (5) onto idler gear shaft (10). - Install gear (10) through the front case bore from the inside. Support the gear and press fit the remaining inner race onto the shaft. - Install holder (9). Install the cap screw and washer. Apply thread lock Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>27.7 ± 3.3 Kgm</td> <td></td> </tr> </table> - Install new O-ring (11) and cover (7). Install the cover cap screws and tighten Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>11 ± 1.2 Kgm</td> <td></td> </tr> </table>	Standard	Actual	27.7 ± 3.3 Kgm		Standard	Actual	11 ± 1.2 Kgm				
Standard	Actual												
27.7 ± 3.3 Kgm													
Standard	Actual												
11 ± 1.2 Kgm													
15	LOWER IDLER GEAR ASSEMBLY	- Press bearing outer race (2) into front housing (1). - Press one inner race (3) onto gear shaft (4). Refer to the match marks for proper bearing assembly. - Insert gear (4) into the front housing bore. Press the remaining inner race (8) onto the shaft.Install plate (5) and snap ring (6). - install nut (7). Apply thread lock Tightening nut <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>35.3 ± 3.9 Kgm</td> <td></td> </tr> </table>	Standard	Actual	35.3 ± 3.9 Kgm								
Standard	Actual												
35.3 ± 3.9 Kgm													
16	IDLER GEAR COVER	- Install a new O-ring (2, Figure 8-65) onto cover (1). Place the cover over the lower idler gear. - Install cap screws (3) and the washers and tighten Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>11 ± 1.2 Kgm</td> <td></td> </tr> </table>	Standard	Actual	11 ± 1.2 Kgm								
Standard	Actual												
11 ± 1.2 Kgm													
17	LOWER PUMP DRIVE ASSEMBLY	- Press bearing outer race (2) into the bore in front housing (1). - Press one inner race (3) of the bearing onto gear shaft (8). Refer to the match marks for proper bearing assembly. - install the gear in the front housing bore and press the remaining inner race (5) onto the shaft. - Install plate (4) and snap ring (7). Apply thread lock Tightening nut <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>3.6 ± 0.4 Kgm</td> <td></td> </tr> </table>	Standard	Actual	3.6 ± 0.4 Kgm								
Standard	Actual												
3.6 ± 0.4 Kgm													

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO								
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NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH GL/QA SKETCH DRAWING OR PHOTHOES								
18	HOUSING ASSEMBLY AND MOUNTING CAP SCREWS	- Install a new O-ring in the groove of the front housing flange. Install new O-rings at the oil transfer holes. Coat the O-rings and mating surfaces with petroleum jelly. - Carefully lower the front housing onto rear housing (2) while meshing the gears. - Apply thread sealer - Install the cap screws and washers. Tighten the cap screws - Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>27.7 ± 3.3 Kgm</td> <td></td> </tr> </table>	Standard	Actual	27.7 ± 3.3 Kgm						
Standard	Actual										
27.7 ± 3.3 Kgm											
	BOSS & BEARING ASSEMBLY	- Press fit bearing (1) to the pump drive boss. - Press fit bearing and boss (2) into cover (1). Install snap ring (3).									
20	PUMP DRIVE GEAR INSTALLATION	- Press bearing inner race (7) onto pump drive gear (5). - Install gear (5) on boss (11). - Apply thread lock - Install the cap screws (1) and tighten - Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>11 ± 1.2 Kgm</td> <td></td> </tr> </table> - Install a new O-ring (14) in cover (12). Coat the O-ring and mating bore in the front housing with petroleum jelly. - Install the pump drive gear assembly into the housing. Install cap screws (13) and the washers and tighten - Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>11 ± 1.2 Kgm</td> <td></td> </tr> </table> - Install a new O-ring (6) onto the cover. - Install the seal in retainer (5). - Apply grease	Standard	Actual	11 ± 1.2 Kgm		Standard	Actual	11 ± 1.2 Kgm		
Standard	Actual										
11 ± 1.2 Kgm											
Standard	Actual										
11 ± 1.2 Kgm											
	INPUT SHAFT	- Coat a new O-ring (2) with petroleum jelly and install it onto retainer (5). - Position retainer (5) over the front housing. Align the hole for dowel pin (3) and install the retainer in the housing. Install snap ring (4). - Heat and install the collar (10). - Coat a new O-ring (9) with petroleum jelly and install it onto input flange (6) and holder (7). Place the assembly into position - Apply thread lock - Install the cap screws (8) and tighten - Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>69.6 ± 3.4 Kgm</td> <td></td> </tr> </table>	Standard	Actual	69.6 ± 3.4 Kgm						
Standard	Actual										
69.6 ± 3.4 Kgm											

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NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES								
22	PUMP CAP SCREW INSTALLATION	- Install the cap screws (2). and washers. Apply thread lock Tightening cap screw <table border="1" style="display: inline-table; margin: 5px;"> <tr><td style="text-align: center;">Standard</td></tr> <tr><td style="text-align: center;">5.4 ± 0.5 Kgm</td></tr> </table> <table border="1" style="display: inline-table; margin: 5px;"> <tr><td style="text-align: center;">Actual</td></tr> <tr><td style="height: 20px;"></td></tr> </table> - Install cap screws (1). Tighten the cap screws Tightening cap screw <table border="1" style="display: inline-table; margin: 5px;"> <tr><td style="text-align: center;">Standard</td></tr> <tr><td style="text-align: center;">11 ± 1.2 Kgm</td></tr> </table> <table border="1" style="display: inline-table; margin: 5px;"> <tr><td style="text-align: center;">Actual</td></tr> <tr><td style="height: 20px;"></td></tr> </table> - Refer to Torque Converter Installation at the beginning of this chapter for information on mounting the torque converter to the transmission.	Standard	5.4 ± 0.5 Kgm	Actual		Standard	11 ± 1.2 Kgm	Actual				
Standard													
5.4 ± 0.5 Kgm													
Actual													
Standard													
11 ± 1.2 Kgm													
Actual													

NOTE

INSPECTION BY

APPROVE BY

MECHANIC

GROUP LEADER (GL)

REV. 00



PLANT REBUILD CENTER

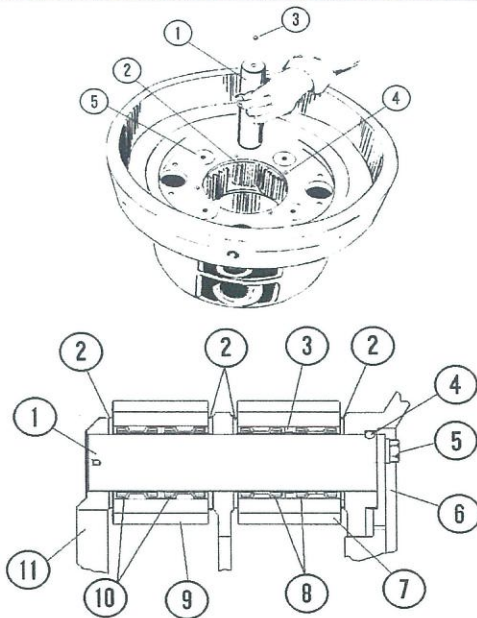
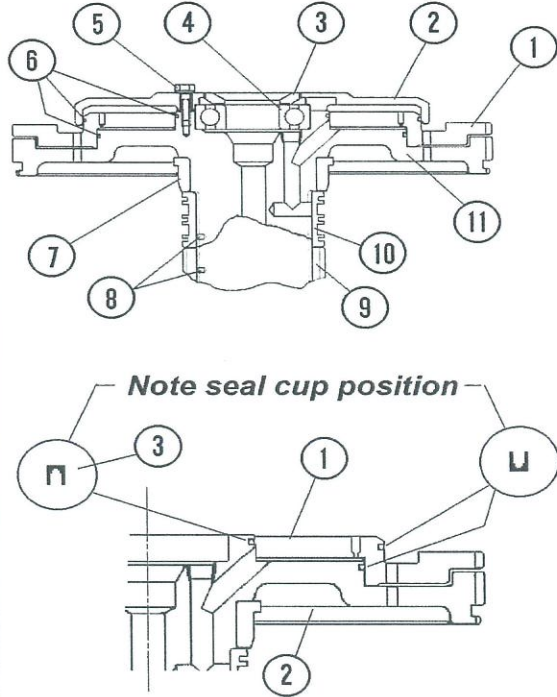
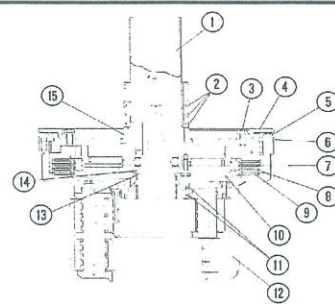
TRANSMISSION ASSEMBLY CHECK SHEET

MACHINE MODEL

HD 1500-7

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POWER TRAIN SECTION

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NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH GL/QA	SKETCH DRAWING OR PHOTHOES				
1	1st & REVERSE CARRIER ASSEMBLY	- If removed, insert sleeves and balls (4), at four locations in the number six carrier. - Position a thrust washer (2), two needle bearings (10), a number five planetary gear (9), and another thrust washer (2) in number five carrier (11) - Install a thrust washer (2) and two needle bearings (8) separated by a collar (3) in the number six planetary gear (7) and another thrust washer (2). Repeat for the remaining three gears. - Install a thrust washer and two needle bearings separated by a collar in the remaining carrier bores. Install the number six planetary pinion gear and another thrust washer - Install cover (6), (5) and install  Apply thread lock  Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>6,5 ± 0,7 kgm</td> <td></td> </tr> </table>	Standard	Actual	6,5 ± 0,7 kgm			
Standard	Actual							
6,5 ± 0,7 kgm								
	OUTPUT SHAFT ASSEMBLY	- Press bearing inner race (7) onto output shaft (11) until fully seated against the shoulder - Drive locating pins (8) into the output shaft holes. - Install collar (10) and worm gear (9) over the locating pins and against the bearing inner race - Press bearing (4) into the output shaft until fully seated against the shoulder of the bore. ⚠ IMPORTANT ⚠ Press against the outer race of the bearing, only. Applying force on the cage or the inner race will damage the bearing. - Coat the piston seals (3) with petroleum jelly. Install the piston seals in the grooves in piston (1) and output shaft (2). - Apply power line oil to the seal contact area. Install piston (1) into the output shaft. - Install cover (2, Figure 9-32). Install cap screws (5) and the washers.  Apply thread lock  Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>6,5 ± 0,7 kgm</td> <td></td> </tr> </table>	Standard	Actual	6,5 ± 0,7 kgm			 Note seal cup position
Standard	Actual							
6,5 ± 0,7 kgm								
3	3rd CLUTCH ASSEMBLY	- Install two seal rings (11) in the grooves in number seven clutch hub (10). Coat the seal rings with petroleum jelly. - Carefully lower the hub into number five carrier assembly (12). - Install a clutch disc (9) over the hub teeth until seated in the carrier. Alternately install clutch plates (8) and the remaining discs. Install wave springs (7) around the discs.						

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NO	PART TO INSTALL OR ASSEMBLY	CONDITION TO BE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTOES										
3	3rd CLUTCH ASSEMBLY	<table border="1"> <tr> <td>Standard discs 4 pcs</td> <td>Actual</td> </tr> <tr> <td>Standard plate 3 pcs</td> <td>Actual</td> </tr> <tr> <td>Standard spring 4 pcs</td> <td>Actual</td> </tr> </table> - Install the O-ring in the hub groove. Install spacer (14). - Install the cap screws (5) and tighten Apply thread lock Tightening cap screw <table border="1"> <tr> <td>Standard 11 ± 1,2 kgm</td> <td>Actual</td> </tr> </table> - Install speed sensor gear (4). Install the cap screws (3) and tighten Apply thread lock Tightening cap screw <table border="1"> <tr> <td>Standard 11 ± 1,2 kgm</td> <td>Actual</td> </tr> </table> - Install seal rings (2) in the collar grooves.	Standard discs 4 pcs	Actual	Standard plate 3 pcs	Actual	Standard spring 4 pcs	Actual	Standard 11 ± 1,2 kgm	Actual	Standard 11 ± 1,2 kgm	Actual			
Standard discs 4 pcs	Actual														
Standard plate 3 pcs	Actual														
Standard spring 4 pcs	Actual														
Standard 11 ± 1,2 kgm	Actual														
Standard 11 ± 1,2 kgm	Actual														
4	REAR CASE ASSEMBLY	- If removed, install collar locating dowel pin (2) into rear case (1). - Press bearing (3) into the case until seated against the shoulder of the bore. - Align collar (4) with the dowel pin and install in the case.													
5	REAR CASE & CARRIER ASSEMBLY	- Install bearing (6) in cover (5). - Coat the outer diameter of seal (7) with sealant. Apply the sealant to the bore in cover (2). Press the seal into the cover. Remove all traces of sealant after installation. - Coat a new O-ring (3) with petroleum jelly. Install the O-ring in the groove of cover (2). - Install cover (2) onto cover (5). - Install cap screws (1) and the washers. Tightening cap screw <table border="1"> <tr> <td>Standard 11 ± 1,2 kgm</td> <td>Actual</td> </tr> </table> - Insert the number six sun gear, the spacer, and the number five sun gear. Insert spacer and bearing (3).	Standard 11 ± 1,2 kgm	Actual											
Standard 11 ± 1,2 kgm	Actual														

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO										
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	FINISH												
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES								
5	REAR CASE & CARRIER ASSEMBLY	- Install a new O-ring (7) in the groove in cover (5). - Place cover assembly (5) over output shaft (4) and align the mounting cap screw holes. - Temporarily install drive flange (1). Tap the flange to install bearing (2) onto the output shaft until fully seated against speedometer drive gear (3). - Remove the drive flange. - Install cap screws (6) and the washers. Tighten to standard torque. Tightening cap screw <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>28,5 ± 1,5 kgm</td> <td></td> </tr> </table> - Install seal ring (9) onto the intermediate shaft. Coat the seal with petroleum jelly to aid in assembly.	Standard	Actual	28,5 ± 1,5 kgm								
Standard	Actual												
28,5 ± 1,5 kgm													
6	REVERSE CLUTCH ASSEMBLY	- Install number six ring gear (2). - Install six guide pins (1). ▲ IMPORTANT ▲ Coat a new clutch piston seal (9) with petroleum jelly. Install the clutch piston seal in the housing groove. Position the seal as shown in the illustration. - Install a new seal on the number six clutch piston. ▲ IMPORTANT ▲ Position the seal with the open part positioned toward the clutch piston oil pressure cavity. - Install 1 clutch disc (6) against piston (3) - Install the separator springs. Continue to alternately install the clutch discs, plates, and separator springs. <table border="1"> <tr> <th>Standard discs</th> <th>Actual</th> </tr> <tr> <td>6 pcs</td> <td></td> </tr> </table> <table border="1"> <tr> <th>Standard plate</th> <th>Actual</th> </tr> <tr> <td>5 pcs</td> <td></td> </tr> </table> - Install 12 piston return springs (4).	Standard discs	Actual	6 pcs		Standard plate	Actual	5 pcs				
Standard discs	Actual												
6 pcs													
Standard plate	Actual												
5 pcs													
7	1st CLUTCH HOUSING ASSEMBLY	- Coat new piston seal rings (3 and 5) with petroleum jelly - Install seal (5) in the clutch housing groove - Install seal (3) in number five clutch piston (2) ▲ IMPORTANT ▲ Position the seal with the cup opening toward the clutch oil pressure cavity - Install the piston in the clutch housing.											

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		UNIT MODEL	PAGE 4 OF 10								
PROCESS	START	ASSEMBLY BY									
	FINISH										
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH GL/QA SKETCH DRAWING OR PHOTHOES								
8	2nd CARRIER ASSEMBLY	- If removed, install four sleeves (4) with detent balls (5). - Position thrust washers (7), needle bearings (10), and a planetary gear in each carrier pinion bore. With the parts aligned, insert frozen shaft (8) and detent ball (9). IMPORTANT Install ring gear (1) as follows: - Insert snap ring (2) in carrier (3). - Align the external teeth of the carrier with the internal teeth of the ring gear. Slide the parts together until the snap ring aligns with the ring gear groove and snaps into place. - Check for proper mating. The ring gear must be stationary on the carrier. - Confirm a gap of approximately 3.2 mm (0.12 in.) exists between the inner diameter of the snap ring and the bottom of the carrier teeth. - Install number four sun gear (13) on the intermediate shaft. Install snap ring (12). - Attach lifting apparatus and a hoist to number four carrier (9). Lower the carrier over the intermediate shaft.									
	BEARING CAGE ASSEMBLY	Assemble the number four carrier retaining cover and bearing cage (11) as follows: - Install bearing (4) in bearing cage (1). - Install snap ring (3). - Install bearing (7) in cover (6). Install seal ring (5). - Assemble the cage to the cover and install snap ring (8). - Install seal rings (2 & 9). - Install the assembled bearing cage on the number four carrier assembly. - Install the cap screws (10) through the access hole in the bearing cage Apply thread lock Tightening cap screw <table border="1"> <tr> <td>Standard</td> <td>Actual</td> </tr> <tr> <td>11 ± 1,2 kgm</td> <td></td> </tr> </table>	Standard	Actual	11 ± 1,2 kgm						
Standard	Actual										
11 ± 1,2 kgm											
10	1st CLUTCH ASSEMBLY	Install the number five clutch plates as follows: - Install plate (6) onto piston (2). - Install a disc, then install a plate. Continue to alternately. Install a separator spring over the guide pins between the discs as they are installed. <table border="1"> <tr> <td>Standard discs</td> <td>Actual</td> </tr> <tr> <td>5 pcs</td> <td></td> </tr> </table> <table border="1"> <tr> <td>Standard plate</td> <td>Actual</td> </tr> <tr> <td>5 pcs</td> <td></td> </tr> </table> - Install piston return springs (8).	Standard discs	Actual	5 pcs		Standard plate	Actual	5 pcs		
Standard discs	Actual										
5 pcs											
Standard plate	Actual										
5 pcs											

ASSEMBLY CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER	DOCUMENT NO	
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NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES																
11	2nd CLUTCH ASSEMBLY	- Install number four clutch housing (3) over the dowel pins and seat against the number five clutch housing. - Install the seals on the number four clutch piston and housing - Install number four ring gear (9). Assemble the number four clutch as follows: - Install plate (5) on piston (1). <table><tr><td>Standard discs</td><td>Actual</td></tr><tr><td>5 pcs</td><td></td></tr></table> <table><tr><td>Standard plate</td><td>Actual</td></tr><tr><td>5 pcs</td><td></td></tr></table> - Install a separator spring over the guide pins between the discs as they are installed. - Install piston return springs (6). - Install housing (8). Install 22 washers (10) and tie-bolts (11) Tightening cap screw <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>38,5 ± 4,2 kgm</td><td></td></tr></table>	Standard discs	Actual	5 pcs		Standard plate	Actual	5 pcs		Standard	Actual	38,5 ± 4,2 kgm								
Standard discs	Actual																				
5 pcs																					
Standard plate	Actual																				
5 pcs																					
Standard	Actual																				
38,5 ± 4,2 kgm																					
12	LOW CLUTCH ASSEMBLY	- Install housing (4) over the dowel pins. - Insert the cap screws (10) through the housing and into bearing cage assembly (11) Apply thread lock Tightening cap screw <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>11 ± 1,2 kgm</td><td></td></tr></table> - Install seal (1) on the housing. Install seal (3) on piston (2). - Install number three ring gear (9). Assemble the number three clutch as follows: - Install plate (5) on the piston. <table><tr><td>Standard discs</td><td>Actual</td></tr><tr><td>4 pcs</td><td></td></tr></table> <table><tr><td>Standard plate</td><td>Actual</td></tr><tr><td>3 pcs</td><td></td></tr></table> ⚠ IMPORTANT ⚠ The ear of the clutch plates for the number three clutch are stamped with the letter "L". - Install a separator spring over the guide pins between the discs as they are installed. - Install piston return springs (8). - Install plate (7) over the dowel pins in the number three clutch housing. Install 17 cap screws and washers (12) Tightening cap screw <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>38,5 ± 4,2 kgm</td><td></td></tr></table>	Standard	Actual	11 ± 1,2 kgm		Standard discs	Actual	4 pcs		Standard plate	Actual	3 pcs		Standard	Actual	38,5 ± 4,2 kgm				
Standard	Actual																				
11 ± 1,2 kgm																					
Standard discs	Actual																				
4 pcs																					
Standard plate	Actual																				
3 pcs																					
Standard	Actual																				
38,5 ± 4,2 kgm																					

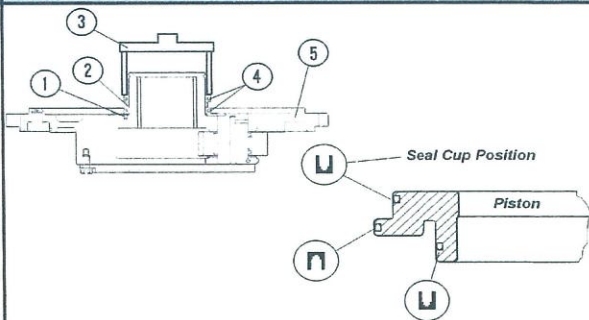
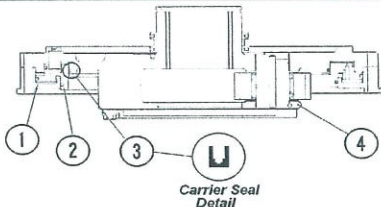
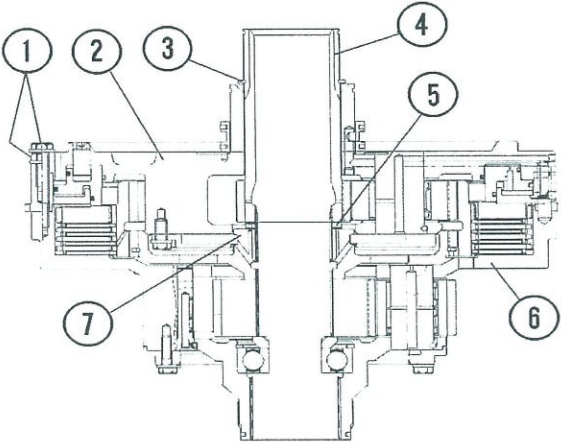
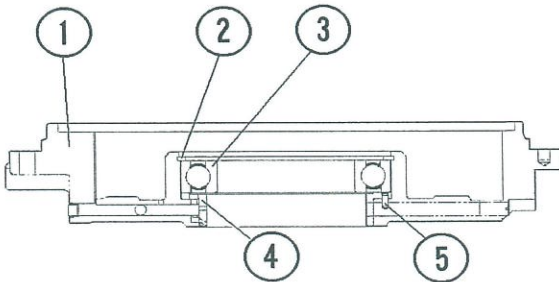
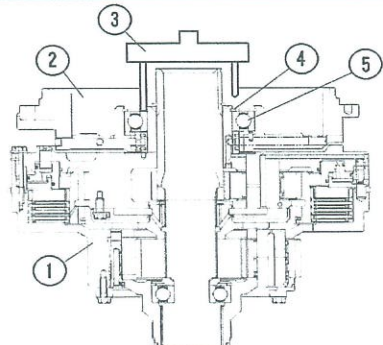
ASSEMBLY CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	6 OF 10

PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES												
13	CLUTCH TEST	Pressurize each clutch through the hydraulic oil port to 3 Bar or more to check for normal operation. <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>3 - 6 Bar</td><td></td></tr></table> Piston stroke each clutch <table><tr><td>3rd Clutch</td><td>2nd Clutch</td></tr><tr><td></td><td></td></tr><tr><td>1st lutch</td><td>Low Clutch</td></tr><tr><td></td><td></td></tr></table>	Standard	Actual	3 - 6 Bar		3rd Clutch	2nd Clutch			1st lutch	Low Clutch					
Standard	Actual																
3 - 6 Bar																	
3rd Clutch	2nd Clutch																
1st lutch	Low Clutch																
14	HIGH CLUTCH RING GEAR INSTALLATION	- Install snap ring (3) in the groove in hub (2). - Align the teeth of ring gear (1) with the teeth on the hub. Press the two parts together until the snap ring fits into the groove on the ring gear. ▲ IMPORTANT ▲ Inspect the snap ring. Ensure a gap of 2.3 mm (0.09 in.) exists between the inside diameter of the snap ring and the hub. - Verify the ring gear is a tight fit on the hub and the parts cannot be separated.															
15	LOW CARRIER & HIGH CLUTCH ASSEMBLY	- Install bearing inner race (15) on number two carrier (1). - Install bearing (11) and snap ring (10). - install sleeves (7) with detent balls. - Install thrust washers (19), needle bearings (20) and gears (16). Align the parts, and install (frozen) shafts (17) and detent balls (18). - Install cover (8). And Install cap screws (9) Apply thread lock Tightening cap screw <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>6,6 ± 0,7 kgm</td><td></td></tr></table> - Install spacer (13), number three sun gear (14), and hub (4). - Install a number two clutch disc, then a plate. <table><tr><td>Standard dics</td><td>Actual</td></tr><tr><td>4 pcs</td><td></td></tr></table> <table><tr><td>Standard plate</td><td>Actual</td></tr><tr><td>3 pcs</td><td></td></tr></table> - Install seal ring (12) in the carrier groove. Apply petroleum jelly to the seal ring and groove.	Standard	Actual	6,6 ± 0,7 kgm		Standard dics	Actual	4 pcs		Standard plate	Actual	3 pcs				
Standard	Actual																
6,6 ± 0,7 kgm																	
Standard dics	Actual																
4 pcs																	
Standard plate	Actual																
3 pcs																	
16	MIDLE CARRIER ASSEMBLY	- Install thrust washers (3), needle bearing (4), and planetary pinion gear (6). Insert shaft (2) in carrier (1). - Install cover (7) and Install the cap screws (8) and tighten Apply thread lock Tightening cap screw <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>6,6 ± 0,7 kgm</td><td></td></tr></table>	Standard	Actual	6,6 ± 0,7 kgm												
Standard	Actual																
6,6 ± 0,7 kgm																	

ASSEMBLY CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
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PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES				
17	COLLAR & PISTON SEAL INSTALLATION	- Using a tool as shown in Figure, align collar (2) with pin (1) and press onto carrier (5). - Coat seal rings (4) and the collar grooves with petroleum jelly. Install the seal rings. - Coat the piston seals (Figure 9-52) with petroleum jelly. Install the piston seals as shown in Figure							
	MIDDLE CARRIER PISTON INSTALLATION	- Coat carrier seal (3) with petroleum jelly. Install the carrier seal. Position the seal cup as shown in Figure - Align the guide hole in piston (2) with the lock pin in carrier (4). Install the piston. Install separator plate (1).							
19	LOW & MIDDLE CARRIER ASSEMBLY	- Install spacer (5) onto hub (7). Install number one sun gear (4) in number three carrier assembly (6). Align the splines in the hub and the spacer. The hole in spacer (5) should be aligned with the hole in the hub (7). - Attach lifting apparatus and a hoist to number one carrier (2). Lower the carrier over the sun gear shaft onto the number three carrier.  Apply thread lock Tightening cap screw <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>11 ± 1,2 kgm</td><td></td></tr></table> - Install snap ring (3).	Standard	Actual	11 ± 1,2 kgm				
Standard	Actual								
11 ± 1,2 kgm									
20	MIDDLE CLUTCH HOUSING BEARING INSTALATION	Install the bearing in the number one clutch housing as follows: - If necessary, install pin (5) in clutch housing (1). - Align bushing (4) with the pin and install in the carrier bore. - Use a bearing installer and a press to install bearing (3). Press the outer race of the bearing into the bore until seated against the bushing. - Install snap ring (2).							
21	HOUSING & CARIER ASSEMBLY	- Using the proper diameter tool (3), press the bearing inner race to assemble number one clutch housing (2) onto number three carrier (1). - After the bearing is seated against the collar, install snap ring (4) in the groove at the bearing inner race.							

ASSEMBLY CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER	DOCUMENT NO
	COMP.MODEL	PUBLIC DATE
	COMP.S/N	REVISION NO
	UNIT MODEL	PAGE 8 OF 10

PROCESS	START	ASSEMBLY BY
	FINISH	

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES												
22	MIDDLE CLUTCH ASSEMBLY	- Install bearing (2) in number one clutch housing (1). Install snap ring (3). - Install hub (4) over the splines on the number one sun gear shaft. - Insert guide pins (2) in the number one housing. Install the number one clutch assembly as follows: - Install a clutch disc on hub(4)Install springs (7) - Install a clutch plate over the disc. - Alternately install the remaining discs,plates, and springs. <table><tr><td>Standard discs</td><td>Actual</td></tr><tr><td>8 pcs</td><td></td></tr></table> <table><tr><td>Standard plate</td><td>Actual</td></tr><tr><td>7 pcs</td><td></td></tr></table> - Coat seal (6) with petroleum jelly. Install the seal on piston (5) with the seal cup positioned as shown in Figure - Install piston return springs (1). - Coat seal (2) with petroleum jelly.Install the seal in clutch housing (1) with the seal cup positioned as shown in the illustration. - Install clutch housing (1). - Install cap screws (9) and tighten Tightening cap screw <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>17,7 ± 2 kgm</td><td></td></tr></table> - Install cover (8) with dowel pin (3) over the sun gear shaft splines. Install snap ring (5). - Install spacer (7) and snap ring (6).	Standard discs	Actual	8 pcs		Standard plate	Actual	7 pcs		Standard	Actual	17,7 ± 2 kgm				Piston Seal Installation Detail Seal Installation Detail
Standard discs	Actual																
8 pcs																	
Standard plate	Actual																
7 pcs																	
Standard	Actual																
17,7 ± 2 kgm																	
23	CASE INSTALLATIONS	- Install new O-rings in the hydraulic oil ports on the face of the rear transmission case. - Attach lifting apparatus and a hoist to the transmission case. Coat a new O-ring (4) with petroleum jelly. Install the O-ring in the groove at the rear of the case. Carefully lower the case over the power train. - Install cap screws (2) and the washers that secure the main case to the rear case. Install the nuts and washers. Tightening cap screw <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>27,5 ± 1,5 kgm</td><td></td></tr></table>	Standard	Actual	27,5 ± 1,5 kgm												
Standard	Actual																
27,5 ± 1,5 kgm																	

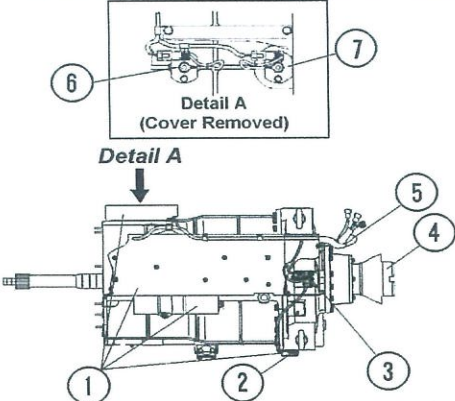
ASSEMBLY CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	9 OF 10

PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES				
24	FRONT PLANETARY & CASE ASSEMBLY	- Install two 8 mm cap screws, plate and gasket (1) and the washers. - Install nine 14 mm cap screws (6) and the washers. Tighten cap screws (1) to standard torque. Tightening cap screw <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>17,6 ± 1,9 kgm</td><td></td></tr></table> - Place cover (4) into position. Install cap screws (3) and the washers.	Standard	Actual	17,6 ± 1,9 kgm				
Standard	Actual								
17,6 ± 1,9 kgm									
25	INPUT SHAFT INSTALLATIONS	- Install two dowel pins (2) and the plates. - Install the cap screws (3) and washers to attach speed sensor gear (4) to transmission input shaft (1) Apply thread lock Tightening cap screw <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>6,5 ± 0,7 kgm</td><td></td></tr></table> - Align the internal splines of sun gear (2), hub (3) and spacer (4). Coat the spacers with petroleum jelly. Center spacer (1) over the bore in bearing (5). ▲ IMPORTANT ▲ Carefully lower the input shaft into the front planetary and clutch assembly. Rotate the shaft as necessary to align the splines. If the alignment is correct, the shaft should drop into place. If the shaft does not drop into place, remove the shaft and realign the splines of the spacer, hub, and sun gear. When the dimension from the top of the speed sensor gear and the face of the case is 6 mm (0.236 in.) or less, the end of the shaft has reached the bearing spacer chamfer ▲ IMPORTANT ▲ When the input shaft is properly seated, the dimension from the face of the transmission case to the face of the speed sensor gear will be approximately 30 mm (1.18 in.)	Standard	Actual	6,5 ± 0,7 kgm				
Standard	Actual								
6,5 ± 0,7 kgm									
26	EXTERNAL COMPONENT	- Install oil pan (1) and a new gasket onto the transmission case. Install the magnet assemblies and new O-rings. Tightening cap screw <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>6,5 ± 0,7 kgm</td><td></td></tr></table> - Install sleeves (2) and new O-rings into the hydraulic oil ports on top of the transmission. - Install O-rings at the ports on valve seat mounting surface (1). - Install the ECMV valve seat with ECMV valves. Install the filter assembly - Install lube oil temperature sensor (2). - Install the speed sensors at input shaft (7), intermediate shaft (6), and output shaft (3).	Standard	Actual	6,5 ± 0,7 kgm				
Standard	Actual								
6,5 ± 0,7 kgm									

ASSEMBLY CHECK SHEET TRANSMISI HD 1500 - 7	WO NUMBER		DOCUMENT NO	
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PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES				
27	SENSOR INSTALLATIONS	● Attach sensor wiring harness (5) and connect the sensor plugs ● Install covers (1) over the sensors, ECMV valves and filter. ● Install output flange (4) onto the output shaft ● Install the cover and a new O-ring. Coat the Oring with petroleum jelly ● Install the cap screw and washer.  Tightening cap screw <table><tr><td>Standard</td><td>Actual</td></tr><tr><td>274,5 ± 29,4 kgm</td><td></td></tr></table>	Standard	Actual	274,5 ± 29,4 kgm				
Standard	Actual								
274,5 ± 29,4 kgm									

NOTE

INSPECTION BY

APPROVE BY

MECHANIC

GROUP LEADER (GL)

CHECKSHEET TESTBENCH TORQFLOW HD 1500-7

DOCUMENT NO	
PUBLICATION	
REVISION	
PAGE	

MACHINE	MODEL		TEST DATE	
	S/N		TEST BY	
T.CONVERTER	MODEL		USED OIL	
	S/N		OIL TEMP	
TRANSMISSION	MODEL		FLOW RATE (Std)	
	S/N		FLOW RATE (Act)	
WO.NO			RPM(Std)	
			RPM(Act)	

NO	RING ITEMS	OIL PRESSURE CHECK		MODULATING TIME		REMARKS
		STD(kg/cm2)	ACT(kg/cm2)	STD	ACTUAL	
1	T/Mission main oil press	37.3 - 42.3				
2	T/M Lubricating	0.2 - 1.0				
3	ECMV Hi	15 - 19		0.2 - 1.4		
4	ECMV Low	18 - 22		0.2 - 1.4		
5	ECMV Medium	17 - 22		0.2 - 1.4		
6	ECMV 3rd	15 - 19		0.2 - 1.4		
7	ECMV 2nd	18 - 22		0.5 - 2.1		
8	ECMV 1st	18 - 22		0.8 - 2.4		
9	ECMV R	18 - 22		1.0 - 2.7		
10	T/C Inlet Oil Pressure	4.9 - 7.4				
11	T/C Lock up Pressure	14.5 - 18.5				
12	T/C Outlet Oil Pressure	3.7 - 6.2				

REMARK:

OIL LEAKAGE

EXCESSIVE HEAT

BREAKAGE / CRACK

ABNORMAL NOISE

☐

YES

☐

NO

☐

YES

☐

NO

☐

YES

☐

NO

☐

YES

☐

NO

NOTE :

TEST BY

APPROVED BY

MEKANIK

QA / GL

PLANT REBUILD CENTER

TORQUE FLOW DELIVERY CHECK SHEET

MACHINE MODEL

HD 1500-7

WO NO:

POWER TRAIN SECTION

DELIVERY CHECK SHEET

TORQFLOW

HD 1500 - 7

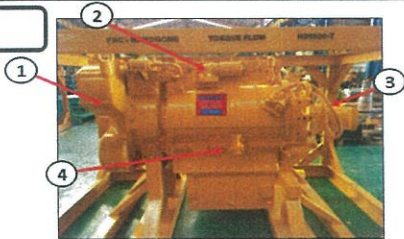
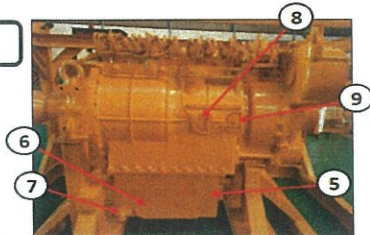
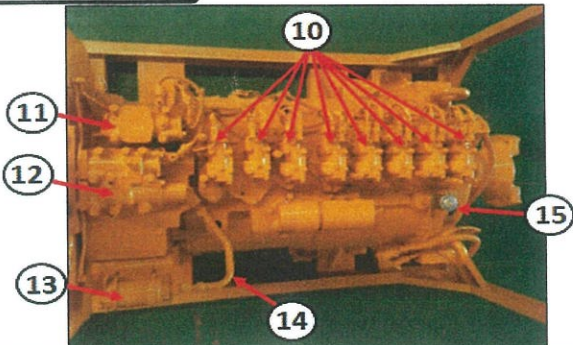
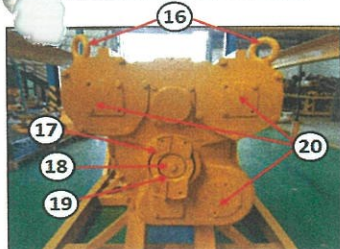
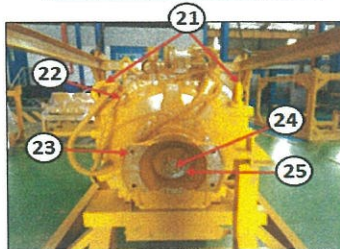
WO. NO		DOCUMENT NO	
COMP.S/N		PUBLIC DATE	
INSPECTION DATE		REVISION NO	
PART NO. COMPONENT	562 -13 -50000	PAGE	

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOOES	NO	ITEM	CHECK	REMARK					
Left Side View 	1	Torque Converter							
	2	Oil Filter Assy							
	3	Wiring Harness							
Right Side View 	4	Lubricating Valve Assy							
	5	Oil Pan							
	6	Magnetic Strainer with Cover							
Top Side View 	7	Drain Oil Plug							
	8	Revolution Sensor (Front)							
	9	Revolution Sensor (Intermediate)							
Front Side View 	10	ECMV Assy (8 pcs)							
	11	T/C Regulator Valve							
	12	Main Relief Valve							
Rear Side View 	13	Lock Up Accumulator							
	14	Lock Up Accumulator Piping							
	15	Breather							
NOTE : PLEASE MARK "✓" FOLLOWING TO THE ACTUAL CONDITION <table><tr><td>OIL LEVEL</td><td>EMPTY OR DRAINED</td><td></td></tr><tr><td>PROPER LEVEL</td><td></td></tr></table>	OIL LEVEL	EMPTY OR DRAINED		PROPER LEVEL		16	Eye Bolt Front (2 pcs)		
	OIL LEVEL	EMPTY OR DRAINED							
	PROPER LEVEL								
	17	Front Coupling							
	18	Bolt Mounting Coupling							
	19	Plate							
	20	PTO (3pcs)							
	21	Eye Bolt Rear (2pcs)							
	22	Name Plate Original							
	23	Rear Coupling							
24	Bolt Mounting Coupling Rear								
25	Plate								
	Painting								
	Stand								
	Masking								
	Check Sheet Delivery								
	Sticker Oil Level								

Note :

INSPECTION BY

APPROVED BY

QA/GL

QA COORDINATOR